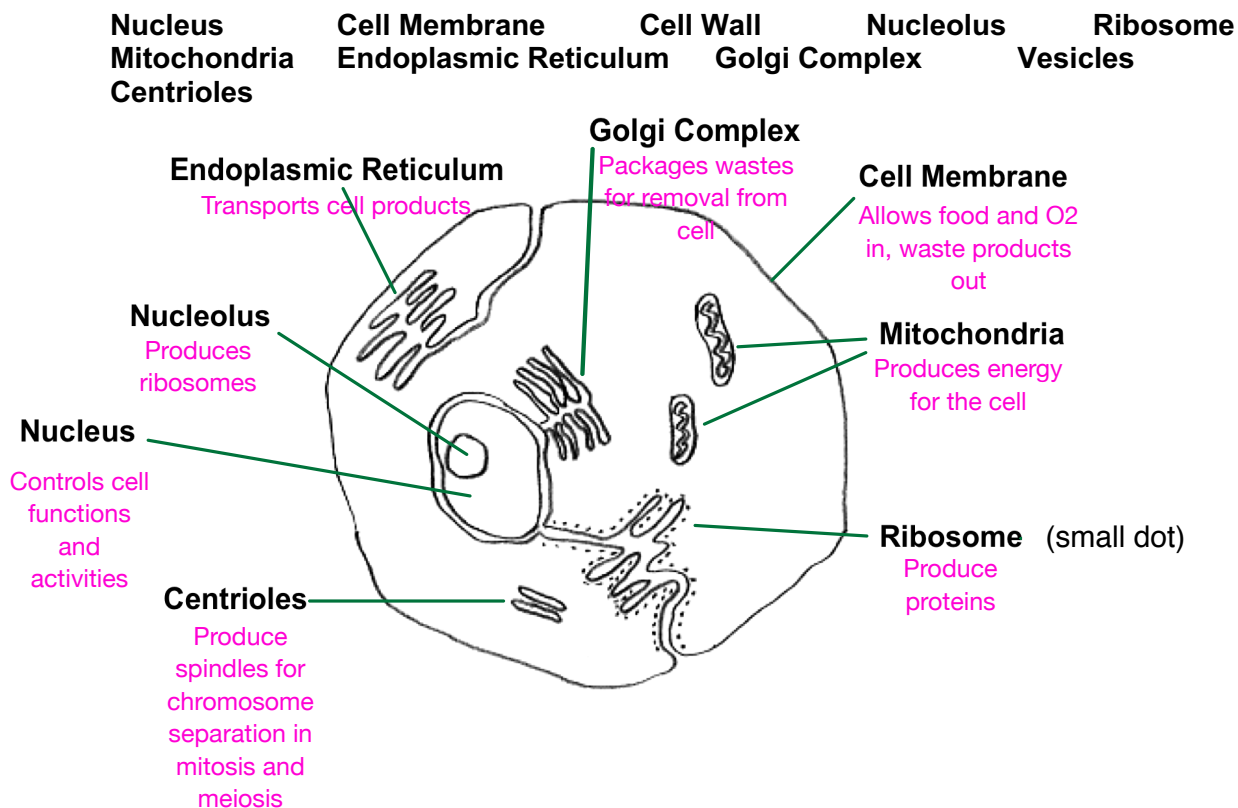


YEAR 10 SCIENCE BIOLOGICAL SCIENCES REVISION

1. Label the following simple cell drawing. For each part, **label its main function**. Choose from the following names - **not all parts are shown in the diagram**.



2. (a) Give a brief description of the DNA molecule.

- Twisted ladder-like structure.
- Double helix of sugar-phosphate.
- Nitrogen bases join them.

- (b) Name the **4 bases** in a DNA molecule.

- Guanine
- Adenin
- Thymine
- Cytosine

- (c) Draw the **base pairs** found in DNA.

G - C A - T

- (d) Draw the adjoining strand of DNA for the following.

- G - T - T - A - C - A - G - T

- C - A - A - T - G - T - C - A

3. (a) Which process of cell division is responsible for the **growth and repair of cells**?

Mitosis

- (b) Which process is involved in producing the **reproductive cells** of an animal?

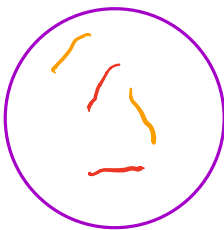
Meiosis

- (c) What is the meaning of **haploid** and **diploid number**?

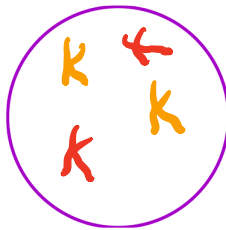
Haploid - half the normal number of chromosomes

Diploid - normal number

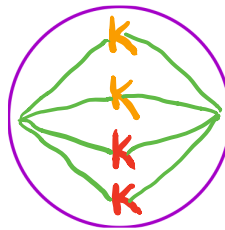
4. An organism has 4 chromosomes in its normal body cells. Draw a series of simple diagrams showing the stages involved in **mitosis**. **Name each stage** and state **what is happening** in each.



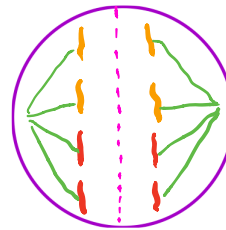
Interphase -
chromosomes
are not visible
(4 are present)



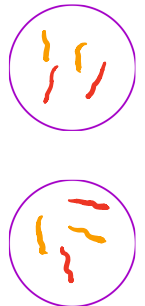
Prophase -
chromosomes
become visible
and duplicate



Metaphase -
chromosomes
line up across the
equator attached
to spindle fibres



Anaphase -
chromosomes are
pulled apart, cell
starts to pinch in
and divide



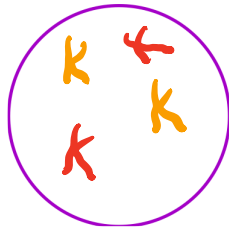
Telophase - cells
divide into two
daughter cells

5. Meiosis occurs in reproductive organs in **two stages**. Draw a series of diagrams to show the stages involved. **Name each stage** and state **what is happening** in each.

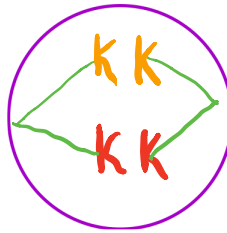
Stage I



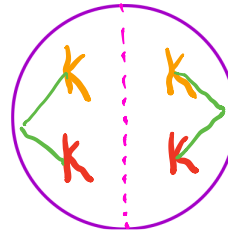
Interphase I - chromosomes are not visible (4 are present)



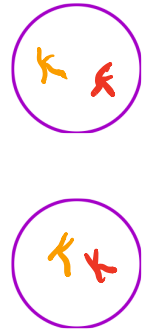
Prophase I - chromosomes become visible and duplicate



Metaphase I - chromosomes line up in pairs across the equator attached to spindle fibres



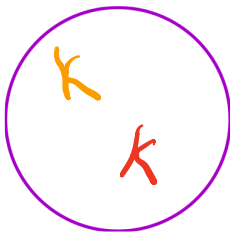
Anaphase I - chromosome pair are pulled apart, cell starts to pinch in and divide.



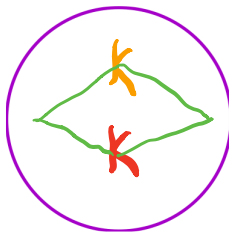
Telophase I - cells divide into two daughter cells with half the normal number of chromosomes

Stage II - proceeds like mitosis

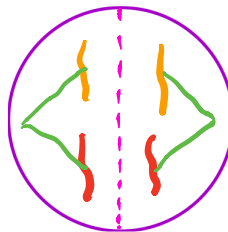
(Only 1 cell shown dividing)



Prophase II - chromosomes are visible



Metaphase II - chromosomes line up across the equator attached to spindle fibres



Anaphase II - chromosomes are pulled apart, cell starts to pinch in and divide



Telophase II - two cells divide into four daughter cells

6. (a) What is the difference between the male and female sex chromosome pair?

XX - female XY - male

- (b) Who determines the sex of a child - the mother or father? Explain.

Father - Y chromosome determines maleness

7. (a) Show, by Punnett square, why half of any particular population would be expected to be female.



- (b) A family has had five girls born into it, and the mother is pregnant with the sixth child. What is the chance that the child will be a girl? Explain why.

50:50 (50%)

Half of male sperm is X, half is Y

8. Briefly describe the significance of the work of Gregor Mendel to our understanding of genetics.

- Grew pea plants.
- Noted that some characteristics dominate others
e.g. Tall dominates dwarf

9. Define the following terms.

allele	One form of a gene e.g. T, t
pure-bred	Alleles the same e.g. TT, tt
hybrid	Alleles different e.g. Tt
homozygous	Alleles the same e.g. TT, tt
heterozygous	Alleles different e.g. Tt
genotype	Genes for a characteristic e.g. TT, Tt, tt
phenotype	How it looks e.g. tall, dwarf
sex-linked	Disorders carried on the X or Y chromosome
dominant	Gene that masks (hides) another e.g. T (tall) masks t (dwarf)
recessive	Gene that is hidden and doesn't show e.g. t (dwarf)
co-dominant	Both genes are dominant and show together e.g. BW - both show as blue and white
incomplete dominance	A new trait shows e.g. RW is pink in some flowers (don't see red or white)
haploid	Half the normal number of chromosomes
diploid.	Normal number of chromosomes

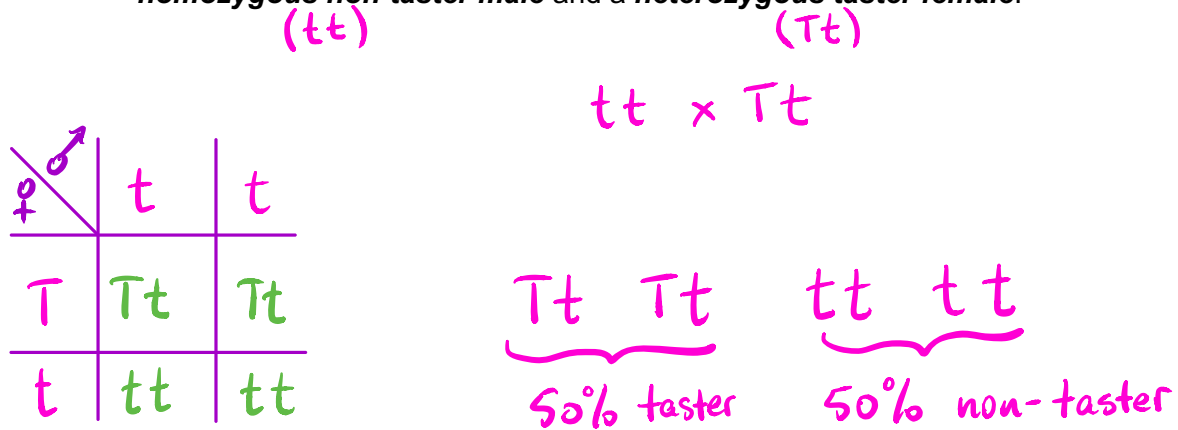
10. Give **one example** of how ***environment*** can affect how a characteristic shows in an individual.

People dye their hair to change its colour.
e.g. Blonde hair (genetic) becomes purple hair (environment).

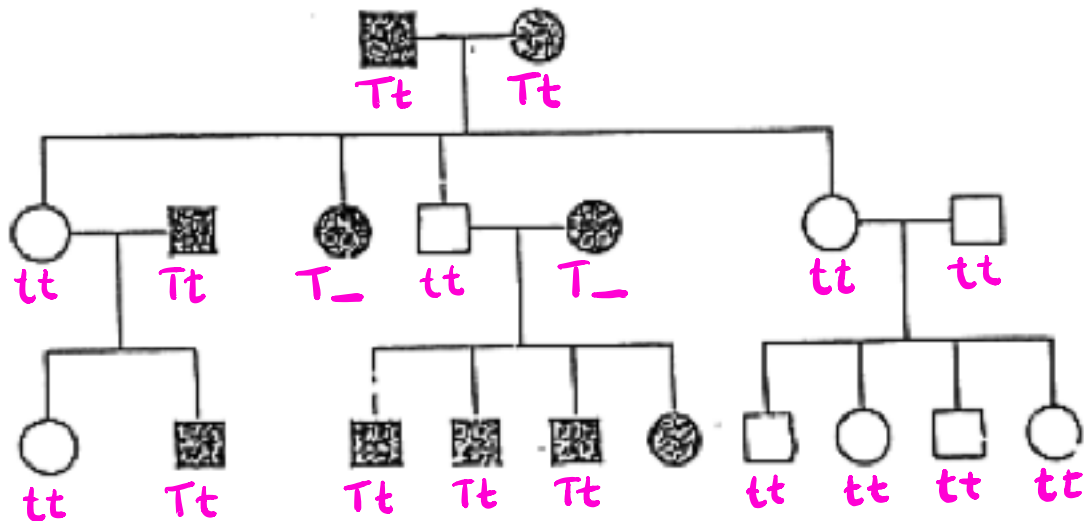
Some wear coloured contacts to change the colour of their eyes.

11. In humans, the ability to taste a chemical phenylthiocarbamide (PTU) is due to the presence of the **dominant gene T**.

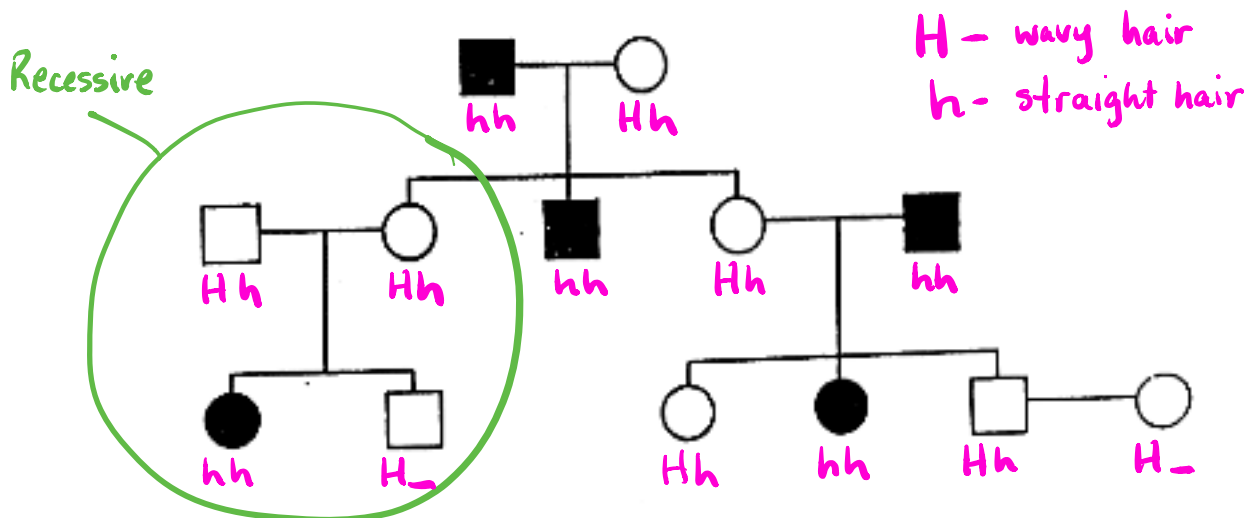
- (a) Determine the genotype and phenotypes of the offspring produced by the marriage of a **homozygous non-taster male** and a **heterozygous taster female**.



- (b) Study the pedigree chart below showing the occurrence of tasters in a family. A shaded square or circle represents a taster. Determine the genotypes of every individual.



12. The pedigree below shows the existence of straight hair in a family.



- (a) Is straight hair a dominant or recessive trait? Explain your answer.

Recessive - two parents don't have the trait but one child does.

- (b) Determine the genotype of all individuals.

13. (a) Explain the difference between **co-dominance** and **incomplete dominance**. You may need to give an example of each to help your explanation.

Both genes are dominant and show together e.g. BW - both show as blue and white

A new trait shows e.g. RW is pink in some flowers (don't see red or white)

- (b) In shorthorn cattle, **red coat colour co-dominates with white coat colour**, producing a roan colour in a heterozygous animal. Determine the F_1 and F_2 generation (first and second generation) offspring for a **red bull** mated with a **white cow**. Set out your working.

$\frac{\text{♀}}{\text{♂}}$	R	R
W	RW	RW
W	RW	RW

(RR) (ww)

$RR \times WW$

RW RW RW RW
100% roan

$\frac{\text{♀}}{\text{♂}}$	R	W
R	RR	RW
W	RW	WW

$RW \times RW$

RR RW RW WW
25% red 50% roan 25% white

- (c) In snapdragon flowers, **red colour is incompletely dominant over white**. Hybrid plants have pink flowers. Determine the genotypes and phenotypes for the offspring produced from a **red flowered** plant with a **pink one**.

♀ \ ♂	R	R
R	RR	RR
W	RW	RW

RR x RW

RR RR RW RW

50% red 50% pink

14. In guinea pigs, **rough coat dominates smooth coat**. Determine the genotypes and phenotypes for the mating of **two rough-coated hybrid** animals.

♀ \ ♂	R	r
R	RR	Rr
r	Rr	rr

Rr x Rr

RR Rr Rr rr

75% rough 25% smooth

15. In humans, suppose that the gene that controls right-handedness is dominant over left-handedness. A heterozygous right-handed male and a left-handed female have four children, three of whom are left-handed.

- (a) If R represents the gene for right-handed, use a Punnett square to determine the expected ratio of children that should be right-handed.

$\frac{\sigma}{\tau}$	R	r
r	Rr	rr
r	Rr	rr

$Rr \times rr$

$Rr \quad Rr$ $rr \quad rr$
 50% right 50% left

- (b) Suggest a reason for difference between the result in part (a) and the actual proportion of children in this family who are right-handed.

Chance - there is a 50/50 of R or r in the sperm of the father.